ANALYZING HEART DISEASE WITH LOGISTIC REGRESSION

MATH 564 – Regression Project

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ABSTRACT

This report presents an analysis of the presence of heart disease using logistic regression. The dataset consists of various demographic and clinical variables, such as age, cholesterol levels, and chest pain types, which are used to predict the likelihood of heart disease. Logistic regression was chosen due to its suitability for binary classification tasks, providing interpretable results in terms of odds ratios. The initial model was refined by addressing issues like influential points, resulting in an improved model fit. The study highlights the significant predictors and discusses the steps taken to ensure a robust and reliable predictive model.

INTRODUCTION

PROBLEM DESCRIPTION

Heart disease is a leading cause of death globally, making early and accurate diagnosis crucial for reducing mortality rates. This project aims to build a predictive model that uses clinical and demographic data to determine the presence of heart disease. By analyzing a set of medical features, the model aims to provide insights into the most critical factors contributing to heart disease.

MOTIVATION

Understanding the key risk factors for heart disease can assist medical professionals in identifying high-risk patients early, allowing for timely intervention. Logistic regression, with its interpretability, was selected to provide clear, actionable insights that can be communicated to medical professionals.

CHALLENGES

Challenges faced during the analysis included handling multicollinearity, identifying and mitigating the impact of influential points, and ensuring model stability. These were addressed through careful diagnostic checks and model refinement.

STRUCTURE OF THE REPORT

The report is organized as follows:

1. Dataset Description and Data Sources
2. Proposed Methodology
3. Analysis and Results
4. Conclusion
5. Bibliography and Credits

DATASET DESCRIPTION AND SOURCE OF DATA

The dataset used in this analysis was obtained from a heart disease study, containing records of clinical and attributes of patients, such as:

- **Age**: Patient's age in years.
- **Sex**: The gender of the patient (typically coded as 1 for male, 0 for female).
- **Chol**: Serum cholesterol level.
- **ChestPain**: A categorical variable indicating the type of chest pain.
- **MaxHR**: Maximum heart rate achieved.
- **Thal**: Classification of Thalassemia condition.
- **RestBP**: Resting blood pressure (in mm Hg) when the patient is at rest.
- **Fbs**: Fasting blood sugar level (> 120 mg/dl means 1, else 0).
- **RestECG**: Resting electrocardiographic results, which indicate the heart's electrical activity at rest.
- **ExAng**: Exercise-induced angina (1 if present, 0 if absent).
- **Oldpeak**: Depression induced by exercise relative to rest (a measure of ST segment depression in ECG, used as an indicator of heart disease).
- **Slope**: The slope of the peak exercise ST segment, which can indicate heart condition severity.
- **Ca**: Number of major vessels colored by fluoroscopy (0-3).
- **AHD**: Presence or absence of coronary artery disease (the target variable in the dataset)

This dataset was pre-processed to handle missing values, standardize numerical variables, and encode categorical features appropriately for logistic regression.

SOURCE OF DATA

https://www.statlearning.com/s/Heart.csv

PROPOSED METHODOLOGY

MODEL SELECTION

Logistic regression was chosen due to its effectiveness in binary classification. The model provides interpretable coefficients that estimate the impact of each predictor on the likelihood of heart disease, expressed as odds ratios.

JUSTIFICATION

Logistic regression is ideal for this task because it:

- Is robust to multicollinearity when regularized.
- Handles binary outcomes effectively.
- Provides clear interpretability of the relationship between predictors and outcomes.

DATA PREPROCESSING

Imputation: Missing values in categorical variables like Thal were imputed with the mode, while numerical variables like Ca were filled using the mean.

Normalization: Numerical variables were standardized to ensure comparable scales.

SELECTION OF THE PREDICTORS

The variables Age, Sex, ChestPain, RestBP, Chol, and MaxHR were chosen because they are well-established predictors of heart disease based on both domain knowledge and statistical considerations. These variables also provide distinct, non-redundant information that helps the model identify the presence of heart disease without overfitting or introducing multicollinearity. The excluded variables, while potentially useful in clinical settings, either introduce redundancy, have weaker predictive power, or may not add significant value in the context of this specific logistic regression model.

DIAGNOSTIC ANALYSIS

To ensure model reliability:

- Multicollinearity was checked using Variance Inflation Factor (VIF).
- Influential Points were identified using Cook's Distance.
- Goodness-of-Fit metrics, including pseudo-R^2, were computed to assess the model's performance.

ANALYSIS OF RESULTS

ASPECT	INITIAL MODEL	REFITTED MODEL
Null Deviance	417.98	389.34
AIC (Akaike Information Criterion)	284.18	187.32
Pseudo R^2	0.363	0.565
Residual Deviance	266.18	169.32
Cook's Distance	Several influential points detected.	Influential points removed, leading to a better fit.

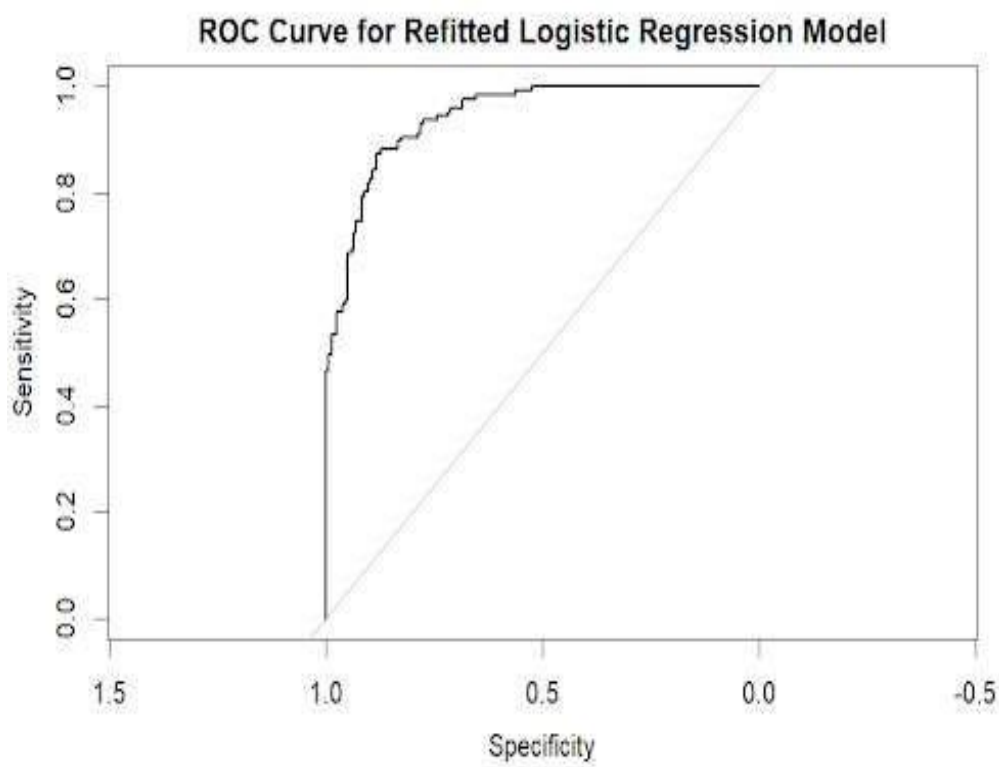
There is a clear improvement in all Pseudo-R^2 metrics after refitting the model. McFadden's R^2 increased from 0.363 to 0.565, indicating a stronger fit of the model to the data.

The residual deviance decreased substantially from 266.18 in the initial model to 169.32 in the refitted model, reflecting a better fit of the refit model to the data.

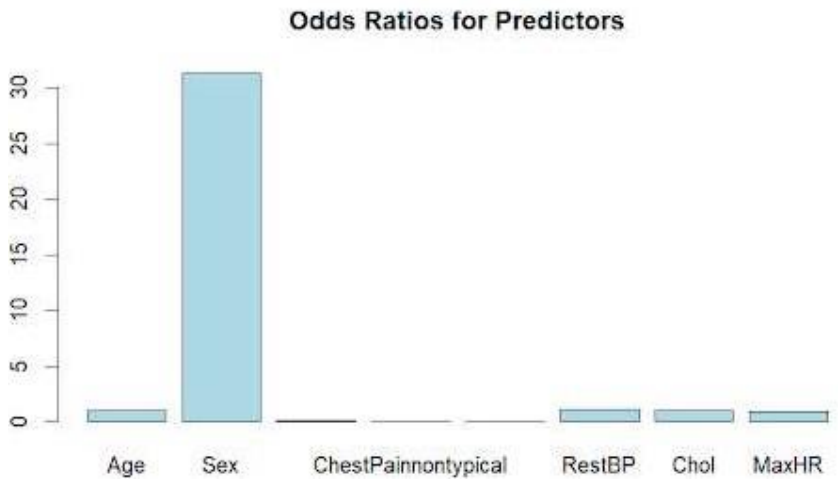
The AIC decreased from 284.18 to 187.32, which confirms that the refitted model is more grasping, meaning it balances goodness of fit with model complexity better than the initial model.

In the initial model, several influential points were detected using Cook's Distance, indicating that certain observations disproportionately affected the model's coefficients. In the refitted model, these influential points were removed, leading to a more stable fit and improving the model's predictive reliability.

VISUALISATIONS



ROC (Receiver Operating Characteristic) curve suggests that the refitted logistic regression model has a reasonably good ability to discriminate between positive and negative cases.



The odds ratio bar plot visually represents the strength and direction of the association between each predictor variable and the outcome (heart disease) in the logistic regression model. The height of each bar corresponds to the odds ratio for that predictor. A bar taller than 1 indicates that an increase in the predictor variable is associated with an increased odds of heart disease. Conversely, a bar shorter than 1 indicates a decreased odds. In this plot, "Sex" appears to be the most significant predictor, with a substantially larger odds ratio compared

to other variables. This suggests that being male is strongly associated with an increased risk of heart disease.

EVALUATION METRICS

pred_class	No	Yes
0	138	19
1	18	108

Confusion matrix, is a common tool used to evaluate the performance of a classification model. In this specific case, it's evaluating a model that predicts the presence or absence of heart disease.

Here's a breakdown of the matrix:

- **True Negative (TN):** 138 instances were correctly predicted as negative (no heart disease).
- **False Positive (FP):** 18 instances were incorrectly predicted as positive (heart disease).
- **False Negative (FN):** 19 instances were incorrectly predicted as negative.
- **True Positive (TP):** 108 instances were correctly predicted as positive.

  Based on this matrix, we can calculate various performance metrics and overall predictive power.

CONCLUSION

KEY FINDINGS

- Logistic regression effectively identified significant predictors of heart disease.
- Removal of influential data points enhanced the model's robustness.
- Autocorrelation was not detected in either mode by the Durbin-Watson test, validating the independence of residuals.
- The refitted model is more robust, as it is less influenced by outliers and exhibits improved predictive accuracy for heart disease.
- This makes the refitted model a preferred choice for practical applications, as it balances accuracy and interpretability, crucial for clinical use.

CHALLENGES ADDRESSED

The project successfully mitigated challenges related to multicollinearity and influential points through data preprocessing and diagnostic evaluation, resulting in a reliable predictive model.

FUTURE DIRECTIONS AND ASPECTS

Future work could include exploring **non-linear models**, such as decision trees or neural networks, which may capture complex relationships between predictors and heart disease risk that logistic regression might not address. Additionally, implementing advanced regularization techniques like **Lasso** or **Ridge Regression** could enhance model performance by optimizing feature selection and reducing overfitting.

Gathering more comprehensive clinical data such as genetic information, lifestyle factors, or longitudinal health metrics could also improve the model's accuracy and robustness. This additional data could provide deeper insights into the risk factors and enhance the predictive capabilities of the model, leading to more reliable clinical decision-making.

This version maintains clarity and focus while providing a bit more detail on potential future directions.

BIBLIOGRAPHY AND CREDITS

https://www.geeksforgeeks.org/heart-disease- prediction-using-logistic-regression-in-r/

https://youtu.be/C4N3_XJJ-jU?feature=shared

https://medium.com/@evelyn.eve.9512/predicting- heart-disease-risk-using-logistic-regression- 7ed58ae2abfd

Gareth James • Daniela Witten •An Introduction to Statistical Learning with Applications in R Second Edition

Link to the dataset

https://www.statlearning.com/s/Heart.csv

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